Learning how to prove Workshop Week 8

Groups

The Fundamental Theorem of Finite Cyclic Groups

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1. In each case determine all the subgroups of the given cyclic group and draw its lattice diagram.
   (a) $G = \langle g \rangle$ with $\mathrm{o}(g) = 8$
   (b) $\mathbb{Z}_{pq}$, where p and q are distinct primes.

Solution:

(a) Consider the mutiplicative cyclic group $C_8 = \langle g \rangle$ of order 8, that is, $\mathrm{o}(g) = 8$. Since the positive divisors of 8 are 1, 2, 4, 8, by the *Fundamental Theorem of Finite Cyclic Groups* we can assert that C_8 has exactly 4 subgroups of orders 1, 2, 4 and 8. It is clear that $\{1\} = \langle 1 \rangle = \langle g^8 \rangle$ and C_8 is the subgroup of order 8. Moreover, for $k \in \{2, 4\}$ we know that $\langle g^{n/k} \rangle$ is the only subgroup of order k. Thus:

$$\begin{aligned} &\text{order 2:} \quad \left\langle g^{8/2} \right\rangle = \langle g^4 \rangle = \{1, g^4\}, \\ &\text{order 4:} \quad \left\langle g^{8/4} \right\rangle = \langle g^2 \rangle = \{1, g^2, (g^2)^2, (g^2)^3\} = \{1, g^2, g^4, g^6\}. \end{aligned}$$

Notice that $\langle g^4 \rangle \subseteq \langle g^2 \rangle$. Using this, it is now easy to draw the lattice diagram:

$$\begin{tikzcd} C_8 \arrow[d, dash] \\ \langle g^2 \rangle \arrow[d, dash] \\ \langle g^4 \rangle \arrow[d, dash] \\ \{1\} \end{tikzcd}$$

(b) The divisors of pq are 1, p, q and pq. Thus the *Fundamental Theorem of Finite Cyclic Group* says that $\mathbb{Z}_{pq}$ has exactly four subgroups of orders 1, p, q and pq. It is clear than the subgroups of $\mathbb{Z}_{pq}$ of orders 1 and pq are $\{\bar{0}\}$ and $\mathbb{Z}_{pq}$, respectively. Moreover, for $k \in \{p, q\}$, we know that $\langle \frac{pq}{k}\bar{1} \rangle$ is the unique subgroup of $\mathbb{Z}_{pq}$ of order k. More precisely:

$$\begin{aligned} \text{order p:} \quad & \left\langle \frac{pq}{p}\bar{1} \right\rangle = \langle \bar{q} \rangle, \\ \text{order q:} \quad & \left\langle \frac{pq}{q}\bar{1} \right\rangle = \langle \bar{p} \rangle. \end{aligned}$$

Notice that $\langle \bar{p} \rangle$ is not contained in $\langle \bar{q} \rangle$, and vice versa.

$$\begin{tikzcd}[arrows=dash] & \mathbb{Z}_{pq} \arrow[dl] \arrow[dr] & \\ p\mathbb{Z}_{pq} \arrow[dr] & & q\mathbb{Z}_{pq} \arrow[dl] \\ & \{\bar{0}\} & \end{tikzcd}$$

2. (a) Find and describe (by listing all the elements) the subgroups of $\mathbb{Z}_{36}$.

   (b) Draw the lattice diagram of $\mathbb{Z}_{36}$.

Solution: (a) and (b). The positive divisors of 36 are 1, 2, 3, 4, 6, 9, 12, 18, 36. Thus, the Fundamental Theorem of Finite Cyclic Groups tells us that $\mathbb{Z}_{36}$ has exactly 9 subgroups, one per positive divisor of 36. Clearly, the subgroups of orders 1 and 36 are $\{\bar{0}\}$ and $\mathbb{Z}_{36}$, respectively. Moreover, for $k \in \{2, 3, 4, 6, 9, 12, 18\}$, applying the Fundamental Theorem of Finite Cyclic Groups we obtain that $\langle \frac{36}{k}\bar{1}\rangle$ is the unique subgroup of $\mathbb{Z}_{36}$ of order k.

Using this, we obtain that

$$\begin{aligned}
&\text{order 2:} && \left\langle \frac{36}{2}\bar{1} \right\rangle = \langle \overline{18} \rangle = \{\bar{0}, \overline{18}\},\\
&\text{order 3:} && \left\langle \frac{36}{3}\bar{1} \right\rangle = \langle \overline{12} \rangle = \{\bar{0}, \overline{12}, \overline{24}\},\\
&\text{order 4:} && \left\langle \frac{36}{4}\bar{1} \right\rangle = \langle \bar{9} \rangle = \{\bar{0}, \bar{9}, \overline{18}, \overline{27}\},\\
&\text{order 6:} && \left\langle \frac{36}{6}\bar{1} \right\rangle = \langle \bar{6} \rangle = \{\bar{0}, \bar{6}, \overline{12}, \overline{18}, \overline{24}, \overline{30}\},\\
&\text{order 9:} && \left\langle \frac{36}{9}\bar{1} \right\rangle = \langle \bar{4} \rangle = \{\bar{0}, \bar{4}, \bar{8}, \overline{12}, \overline{16}, \overline{20}, \overline{24}, \overline{28}, \overline{32}\},\\
&\text{order 12:} && \left\langle \frac{36}{12}\bar{1} \right\rangle = \langle \bar{3} \rangle = \{\bar{0}, \bar{3}, \bar{6}, \bar{9}, \overline{12}, \overline{15}, \overline{18}, \overline{21}, \overline{24}, \overline{27}, \overline{30}, \overline{33}\},\\
&\text{order 18:} && \left\langle \frac{36}{18}\bar{1} \right\rangle = \langle \bar{2} \rangle = \{\bar{0}, \bar{2}, \bar{4}, \bar{6}, \bar{8}, \overline{10}, \overline{12}, \overline{14}, \overline{16}, \overline{18}, \overline{20}, \overline{22}, \overline{24}, \overline{26}, \overline{28}, \overline{30}, \overline{32}, \overline{34}\}.
\end{aligned}$$

(b) Notice that

$$\begin{aligned}
&\langle \overline{18} \rangle \subseteq \langle \bar{9} \rangle, \quad \langle \overline{18} \rangle \subseteq \langle \bar{6} \rangle, \quad \langle \overline{18} \rangle \subseteq \langle \bar{3} \rangle \text{ and } \langle \overline{18} \rangle \subseteq \langle \bar{2} \rangle,\\
&\langle \overline{12} \rangle \subseteq \langle \bar{6} \rangle, \quad \langle \overline{12} \rangle \subseteq \langle \bar{3} \rangle, \quad \langle \overline{12} \rangle \subseteq \langle \bar{4} \rangle \text{ and } \langle \overline{12} \rangle \subseteq \langle \bar{2} \rangle,\\
&\langle \bar{9} \rangle \subseteq \langle \bar{3} \rangle,\\
&\langle \bar{6} \rangle \subseteq \langle \bar{3} \rangle \text{ and } \langle \bar{6} \rangle \subseteq \langle \bar{2} \rangle,\\
&\langle \bar{4} \rangle \subseteq \langle \bar{2} \rangle.
\end{aligned}$$

Using these we get the following lattice diagram:

$$
\begin{tikzcd}
 & & \mathbb{Z}_{36} \arrow[dl, dash] \arrow[dr, dash] & & \\
 & \langle\bar{2}\rangle \arrow[dl, dash] \arrow[dr, dash] & & \langle\bar{3}\rangle \arrow[dl, dash] \arrow[dr, dash] & \\
\langle\bar{4}\rangle \arrow[dr, dash] & & \langle\bar{6}\rangle \arrow[dl, dash] \arrow[dr, dash] & & \langle\bar{9}\rangle \arrow[dl, dash] \\
 & \langle\overline{12}\rangle \arrow[dr, dash] & & \langle\overline{18}\rangle \arrow[dl, dash] & \\
 & & \{\bar{0}\} & &
\end{tikzcd}
$$